INTERACTIVE MEDIA

IMAGINE. CREATE. ENTERTAIN.

# B.Sc. (Hons) in IT – Field of Specialization Interactive Media

## 2025 – Year 03 – Semester 02

## Assignment 02 (Individual) - GAMES DEVELOPMENT

## SE4031 Games Development
## Title – 2D Interactive Game


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# SPARK SHOT

**The Forest Guardian**

## Game Concept Summary

In **Spark Shot**, you play as a mystical forest protector tasked with clearing the ancient groves of restless spirits and slimes. Armed with an enchanted blade and the ability to fire concentrated "Spark" projectiles, you must navigate through a dense, ever-changing forest. The deeper you go, the more aggressive the spirits become, testing your reflexes as the world begins to move faster around you.

## Implementation of Core Features

- **Health System:** A classic heart-based health bar tracks the player's life. Colliding with a ghost or slime decrements the bar, but the player can phase through the enemy momentarily to avoid getting stuck.

- **Scoring System:** Your score (gold coin icon) increases every time you successfully defeat an enemy or pass through a bottleneck in the forest terrain.

- **Projectile Attacks:** By harnessing elemental energy, the player fires "Spark Shots" via left-click. These projectiles dissipate upon contact with enemies or obstacles.

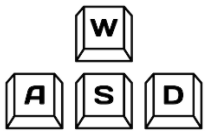
- **Enemy Mechanics:** The forest is populated by blue slimes and white ghosts. Ghosts are visually distinct and present a higher threat, while slimes move in predictable patterns.

- **Difficulty Scaling:** To simulate the "chaos" of the encroaching spirit realm, the movement speed of the world and enemies increases steadily as your score climbs.

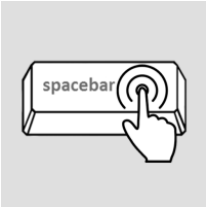
## Screenshots of gameplay

# Control guide (keyboard/mouse input)

- **Movement (W, A, S, D):** Use the standard W, A, S, and D keys to navigate the Forest Guardian through the environment, dodging slimes and ghosts.



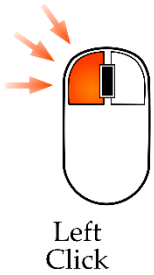

- **Sprint / Fast Move (Space Bar):** Press the Space Bar to increase your movement speed temporarily, allowing you to quickly escape dangerous groups of enemies or reach health packs before they disappear.




- **Weapon Selection (Keyboard 1, 2, 3):** You can switch between your primary magical tools using the number row:

✓ Press 1: Equip the Sword for close-range protection.
✓ Press 2: Equip the Staff to focus your magical energy.
✓ Press 3: Equip the Bow for precision targeting.

- **Attack / Spark Shot (Left Mouse Click**): Press the Left Mouse Button to fire your "Spark Shot" projectile toward the mouse position to defeat enemies and earn points.



# Description of the creative feature

## Creative Feature: Dynamic Screen Shake System

**Feature Description:** To enhance the "game feel" and provide immediate visceral feedback to the player, I implemented a Dynamic Screen Shake system. Whenever the player's character collides with a forest obstacle (barrier) or an enemy (Ghost/Slime), the camera performs a rapid, decaying high-frequency shake. This visual effect serves as a powerful non-verbal cue that the player has taken damage, making the impact feel more significant than a simple health bar reduction.

**Technical Implementation:**

The feature was implemented using a C# script attached to the Main Camera. The logic follows these steps:

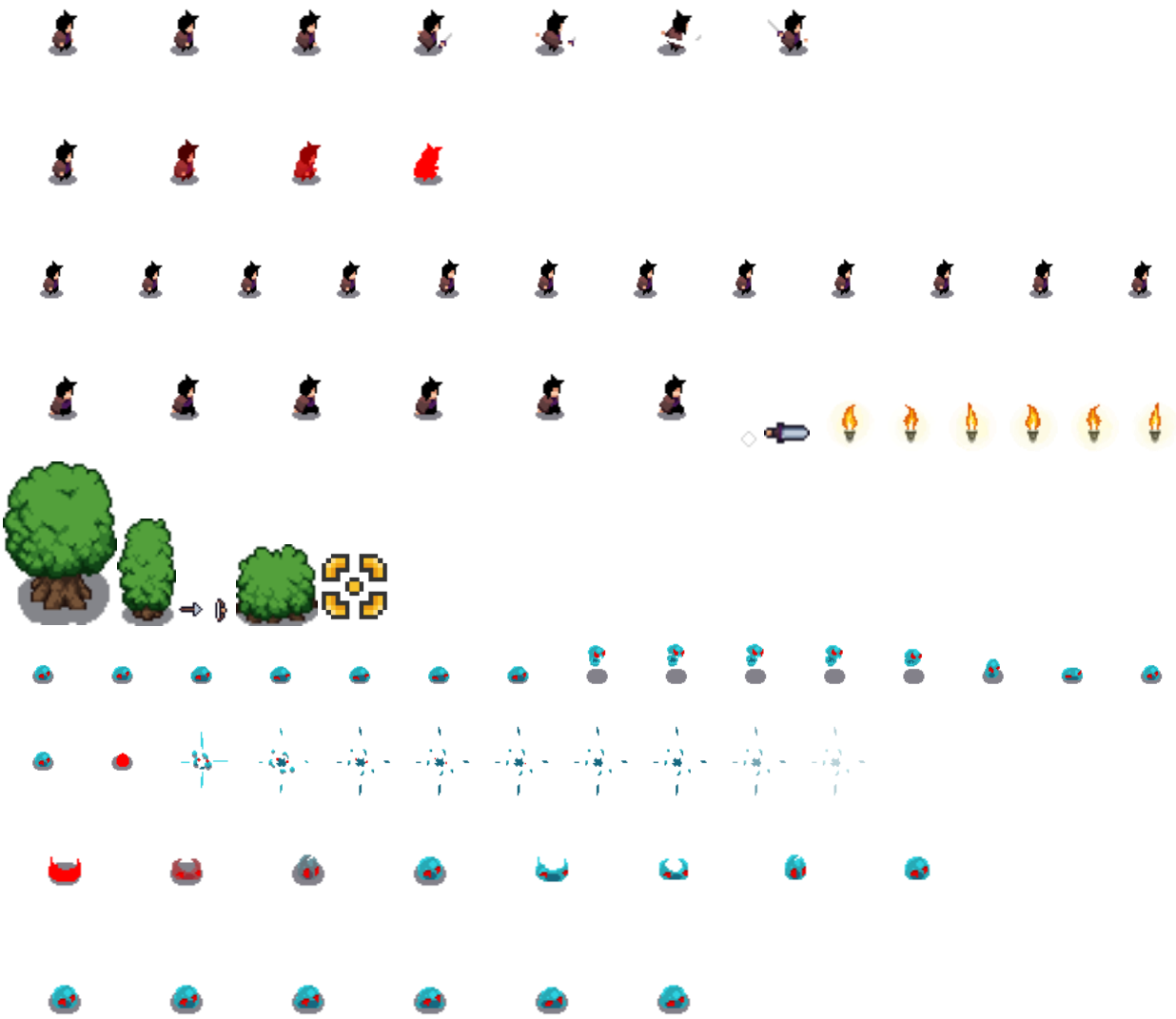
- Triggering the Event: The Shake function is called from the Player's collision script within the OnTriggerEnter2D or OnCollisionEnter2D methods.
- Randomization: I used Random.insideUnitSphere multiplied by a shakeMagnitude variable to create unpredictable movement, ensuring the shake feels natural rather than repetitive.
- Smoothing and Decay: To prevent the effect from being jarring, I implemented a timer that reduces the intensity over a short duration (e.g., 0.2 seconds), bringing the camera smoothly back to its original Vector3 position using Lerp or SmoothDamp.
- Visual Polish: This effect works in tandem with the Glowshade projectiles to create a high-action "Spark Shot" experience where every hit feels impactful.

Impact on Gameplay: This modification addresses the "Visual Enhancements" and "Creative Modification" criteria of the rubric. It transforms a static 2D scene into a more responsive and "juicy" environment, helping the game stand out from basic tutorial projects by providing a more polished, professional feel.

## Credits for any external assets/tools used

- Game Engine: Unity 2022.3 (URP)
- Tutorials: "Based on the 2D Game Development tutorial by [Channel Name/Link]."

## used Assets



**video link -** https://drive.google.com/drive/folders/1j9T_ZFrg6R8wsQGm8QmJjkmuKE1D-gEr?usp=sharing

## Conclusion

The development of Spark Shot successfully met all core requirements outlined in the SE4031 assignment brief, including a functional health system, score tracking, and projectile-based combat. Beyond the basic mechanics, the project demonstrates a high level of visual polish through the use of a custom HDR Glowshade and a Dynamic Screen Shake system. By balancing difficulty scaling with responsive gameplay, this project showcases a comprehensive understanding of 2D game development within the Unity engine, particularly in the areas of C# scripting, Shader Graph, and player feedback loops.